

ARTIFICIAL INTELLIGENCE DIVISION

GROUP PROJECT 3

Robot Fault Diagnosis through Transfer Learning and Digital Twins

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1 Introduction

In the context of Industry 4.0, industrial robotic systems are subject to increasing operational constraints in terms of availability and reliability. Predictive diagnosis and smart maintenance therefore represent strategic levers for optimising performance and reducing operating costs.

1.1 Industrial context and challenges

Modern robotic equipment embeds multiple actuators, sensors and sophisticated control systems where the smallest malfunction can lead to significant economic consequences. Unplanned production downtime is estimated to cost large manufacturing companies several million euros per year.

Artificial intelligence (AI) and machine learning (ML) offer promising prospects for revolutionising traditional maintenance approaches. Automatic fault diagnosis based on time-series analysis is a particularly relevant avenue of innovation.

1.2 Client presentation

The client is Dr Zhiguo Zeng, a researcher at the Industrial Engineering Laboratory (LGI) of CentraleSupélec. His research focuses on system reliability and predictive maintenance, with an emphasis on applying AI techniques to industrial problems.

1.3 Industrial problem and technical challenges

In the industrial sector, the motors that make up robots are exposed to various faults: complete actuator blockage, progressive angular inaccuracy, or positioning errors. The presence of many interconnected motors and the diversity of possible failures make diagnosis particularly complex.

Downtime is a critical economic issue, encompassing production loss, the mobilisation of specialised technicians and delivery delays.

The core technical challenge lies in the ability to detect and classify faults affecting a robotic arm (illustrated below) while limiting the amount of required data. The goal is to determine the motors' health solely from the analysis of their dynamic response to a given command, without access to internal information (voltage, temperature, encoders).

This “black-box” approach is simple to implement but raises algorithmic challenges in extracting discriminative features.

1.4 Project objectives

An initial algorithm developed in MATLAB by the client achieves an accuracy of 86%. Within the scope of this Artificial Intelligence Division project, our dual objective is:

- **Performance improvement:** Significantly surpass the 86% accuracy by exploring advanced deep-learning architectures.
- **Solution modernisation:** Provide a Python implementation to facilitate integration and future evolution.

Our approach relies on transfer learning and domain adaptation techniques, leveraging simulated data to improve performance on real data. The project aims to demonstrate the relevance of adversarial networks for robotic fault diagnosis.

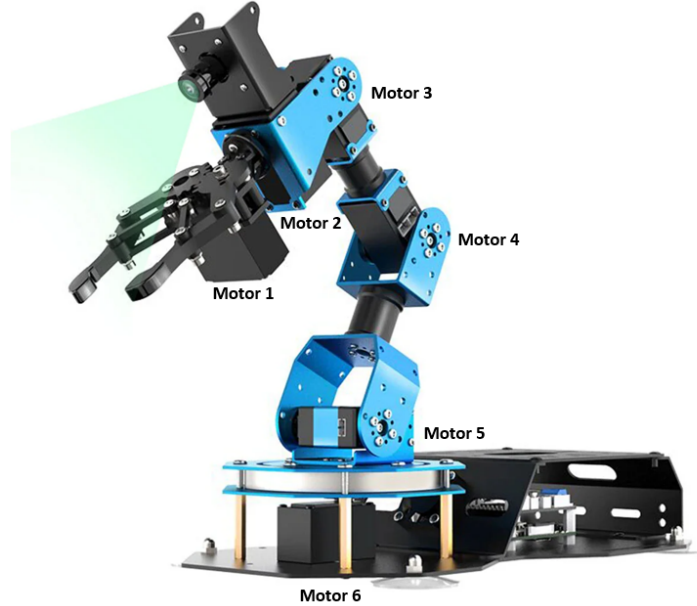


Figure 1: Six-degree-of-freedom industrial robot

2 State of the Art

2.1 Context and issues

With the rise of Industry 4.0, predictive maintenance has emerged as a strategic solution for reducing unplanned shutdowns, extending equipment lifetime and optimising operating costs. Among the flagship tools of this transformation, the digital twin occupies a central position: it digitally reproduces a physical system and tracks its behaviour in real time to anticipate potential failures [1]. Nevertheless, several major obstacles hinder large-scale deployment of these technologies, including modelling complexity, data heterogeneity, IT infrastructure limitations, and the lack of interpretability of AI-generated decisions [1,2].

2.2 Existing approaches and techniques

Various strategies have been developed to reinforce the accuracy of digital twins for predictive maintenance:

- **Physics-based modelling:** relies on mathematical models replicating system behaviour. Highly reliable but requires deep domain expertise [1].
- **Machine learning and deep learning:** AI, through CNNs, RNNs or Transformer architectures, detects fault patterns from time series, images or sensor signals [3].

- **Multimodal data fusion:** models gain accuracy when integrating various information sources (temperature, vibration, maintenance logs, thermal images), compensating limitations of a single sensor or signal [4].
- **Hybrid predictive maintenance:** combines physics-based models and neural networks to harness the robustness of equations and the flexibility of AI [5].
- **Model interpretability:** explainability methods such as SHAP or LIME are increasingly used to clarify model decisions and build industrial users' trust [4,6].
- **Transfer learning:** reuses pretrained models from other machines or contexts to overcome data scarcity, reducing the need for domain-specific data [2,5].

2.3 Current limitations and challenges

Adopting digital twins for predictive maintenance faces several hurdles:

- **Lack of high-quality data:** some machines are not equipped with advanced sensors or rarely fail, making learning difficult. Class imbalance (few failures vs many normal samples) compromises model robustness [1,2].
- **High computational cost:** digital twins require significant compute and storage resources, especially with complex AI models and real-time data streams [1].
- **Limited generalisation:** models developed for a specific machine or site often struggle to adapt elsewhere without fine-tuning or retraining. Few-shot learning and model optimisation aim to address this [2,5].
- **Poor interpretability:** complex algorithmic decisions remain hard to interpret for field engineers, hampering acceptance in critical environments [4,6].
- **Sustainability and energy efficiency:** deploying intelligent digital twins must also consider their energy footprint, particularly for embedded or edge AI systems [7].

2.4 Perspectives and future work

Research is moving towards digital twins that are:

- **More interpretable,** via systematic incorporation of XAI techniques into AI architectures [4,6].
- **More generic and adaptable,** exploiting transfer learning and simulation to limit reliance on real data [2,5].
- **More eco-efficient,** through lightweight models (TinyML) and edge computing to curb energy needs [7].
- **Better connected,** integrated into interoperable IoT ecosystems to facilitate data sharing between heterogeneous equipment [1].

2.5 Conclusion

Digital twins combined with AI are revolutionising industrial maintenance by enabling more accurate diagnosis, proactive interventions and a better understanding of machine behaviour. Nevertheless, to become standard practice, efforts are still needed regarding data quality, model generalisation, result readability and system energy optimisation. By combining physics-based modelling, AI, explainability and energy efficiency, the next generation of digital twins can meet the growing demands of the factory of the future.

3 Description of the Work Performed

3.1 Data study and augmentation

3.1.1 Dataset analysis

Analysis of the dataset revealed significant differences between simulated and real trajectories. Computing the cosine similarity between each simulated trajectory and the real trajectories of its respective class, and keeping the maximum value, yielded a normal distribution centred around 0.7, with many simulations showing very low similarity to real trajectories.

3.1.2 Data augmentation

To remedy this issue, we assumed that point-wise differences between real and simulated trajectories follow a Gaussian noise whose means and standard deviations we estimated empirically. For each trajectory class, we selected the three most faithful simulations and replicated them roughly one hundred times, adding noise tensors drawn from normal laws with the same means and standard deviations at each time step. This made them predominant in the dataset while preventing models from over-fitting these trajectories. Maintaining the same number of trajectories per class was essential, as it was difficult to estimate how to balance classes to reflect a robot arm’s real life-cycle.

3.2 LSTM

3.2.1 Proposed architecture

The LSTM was the first model we explored. It is also the simplest and thus serves as a baseline for more elaborate models.

Model layers

- Two unidirectional LSTM layers with hidden size 105 and dropout 0.1.
- One linear layer 105×100 .
- One batch-normalisation layer.
- One ReLU activation layer.
- One linear layer 100×9 .

The resulting vector is
emphnot passed through a softmax layer, as the multi-class cross-entropy loss handles that.

3.2.2 Evaluation and performance

Training the model for 90 epochs yielded 68% accuracy, surpassing the 61.56% achieved by the client. However, the more relevant metric is the F1-score:

F1-score

- Healthy: 0.0645
- Motor 1 Steady State: 0.8889
- Motor 1 Stuck: 0.9412
- Motor 2 Steady State: 0.8235
- Motor 2 Stuck: 0.6923
- Motor 3 Steady State: 0.9091
- Motor 3 Stuck: 0.8000
- Motor 4 Steady State: 0.6957
- Motor 4 Stuck: 0.7500

These figures show a marked improvement on all trajectories except the healthy ones.

3.2.3 Potential developments

It appears that the model, naturally good at classifying healthy trajectories, sees its performance on them degrade when trained on the unmodified dataset, while the opposite occurs on all other trajectories. Augmenting with data generated from only three trajectories is not rich enough for proper generalisation and only benefits tasks where the model originally performs poorly. Thus, a compromise likely exists for each class between the number of faithful trajectories to select for augmentation and the minimum quality to require. These results remain promising within a “small-model” optimisation approach and for future, more complex models built upon the LSTM.

3.3 DANN (Domain-Adversarial Neural Networks)

This section presents our main contribution: developing an optimised DANN architecture for robotic fault diagnosis.

3.3.1 Proposed architecture

Our DANN model forms the core approach developed for this project. The architecture includes:

Feature extractor

- 1-D convolution (64 filters, kernel 7) + BatchNorm + ReLU
- 4 ResBlocks with increasing filter counts (64, 128, 256, 512)
- Squeeze-and-Excitation attention adapted to 1-D signals
- GlobalAveragePooling for length invariance

Adversarial components

- Task classifier: dense architecture (512→256→128→6 neurons)
- Domain discriminator with a Gradient Reversal Layer
- Composite loss: $L_{\text{total}} = L_{\text{class}} + \lambda L_{\text{domain}} + \mu L_{\text{reg}}$

3.3.2 Evaluation and performance

Experimental protocol DANN evaluation followed a rigorous protocol with five independent runs to ensure reproducibility. The dataset comprised:

Simulated training data:

- 400 samples per class for nine fault classes
- 80/20 split for training/validation
- 2,880 training samples, 720 validation samples

Real data for transfer:

- 10 samples per class (90 total)
- 30% used for domain adaptation (27 samples)
- 70% used for final evaluation (63 samples)

Results Average final performance (five runs):

- Mean accuracy: $93.97\% \pm 2.54\%$
- Best training accuracy: $97.78\% \pm 0.78\%$

Per-run detail:

1. Run 1: 93.65%
2. Run 2: 96.83%
3. Run 3: 92.06%
4. Run 4: 96.83%
5. Run 5: 90.48%

Robustness analysis The observed variance ($\pm 2.54\%$) across runs is acceptable for this type of industrial application, confirming the approach’s reliability for practical deployment.

Training observations The gap between best training accuracy ($97.78\% \pm 0.78\%$) and average final accuracy ($93.97\% \pm 2.54\%$) suggests that early stopping could have been beneficial, potentially retaining performance near the observed peak while preventing slight over-fitting.

3.4 GTN (Graph Transformer Network)

This section details experimentation with the GTN architecture for robotic fault diagnosis, as a comparison with the DANN approach.

3.4.1 Proposed architecture

The GTN model is based on a Transformer adapted to 1-D signals, using the residue (difference between command and motor response) as input vector.

Input pipeline

- Combined input: command + response $\rightarrow$ residue computation
- 1-D convolution (input dim 3 $\rightarrow d$)
- Conv block + BatchNorm + GELU

Transformer block

- N EGT (Enhanced Graph Transformer) blocks
- Attention mechanism tailored to time-series processing

Classification

- Adaptive 1-D pooling for temporal invariance
- Dense MLP with residuals + dropout
- Softmax output over $C = 9$ classes

Augmentation strategy

- Gaussian noise added to inputs
- $\sigma = 0.001$ (0.1% of amplitude)
- Dataset doubled (one noisy copy per sample)
- Applied only during training

Transfer learning GTNResidu was first trained on simulated data. During transfer, certain layers are retained:

- Temporal convolution layers
- Initial projection layers
- First Transformer blocks

Final layers (later Transformer blocks, MLP projection and classifier) are re-initialised and retrained with a low learning rate. This leverages generic representations learned from simulation while adapting the model to real data. The GTNResidu’s hierarchical architecture allows gradual adaptation, retaining low-level patterns and relearning decision functions.

3.4.2 Evaluation and performance

Experimental protocol GTN was evaluated under the same protocol as DANN, with transfer learning on a small dataset. Training used simulated data from the digital twin. Evaluation relied on transfer learning with or without augmentation.

Evaluation conditions:

- Simulated dataset identical to that used for DANN
- Different proportions from DANN: 80% of real data for training, 20% for validation; 50% of the test set for transfer learning and 50% for testing.

Results Average test accuracy:

- **73.33%** with transfer learning *and* augmentation
- **65.56%** with pretrained model without adaptation
- **62.22%** with transfer learning *without* augmentation
- **55.56%** with neither adaptation nor augmentation

These results show a net performance gain thanks to transfer learning, although combining it with augmentation was not beneficial in every case.

Method	Accuracy	Gap vs DANN
GTN (best config.)	73.33%	−20.64 points
DANN (average)	93.97%	Reference

Table 1: Performance comparison between DANN and GTN

Comparison with DANN Despite its originality, the GTN approach achieves markedly lower performance than DANN.

Performance analysis GTN performance is limited by:

- A dataset too small to train such an architecture effectively
- Strong computational constraints (model cannot be trained locally)
- Greater sensitivity to over-fitting and data variability

Observations GTN shows theoretical potential, but experimental conditions did not allow its full capacity to be exploited. Combining larger datasets and a powerful computing platform (HPC) will be necessary to validate its effectiveness in a real industrial context.

4 Conclusion and Perspectives

4.1 Project hindsight and lessons learned

This robotic fault-diagnosis project confronted us with the complexity of real industrial issues. The experience taught us the importance of simulated-to-real adaptation, the core challenge of industrial transfer learning.

4.1.1 Methodological insights

The rigorous scientific approach imposed by Dr Zeng underscored the importance of experimental reproducibility. The five independent runs of our DANN model revealed acceptable variance ($\pm 2.54\%$), confirming the robustness of our approach.

The gap between training (97.78%) and final (93.97%) performance highlighted the need for continuous over-fitting monitoring and the value of techniques such as early stopping.

4.1.2 Technical challenges overcome

Implementing adversarial networks proved more complex than anticipated, requiring in-depth understanding of gradient reversal mechanisms and adversarial optimisation. This difficulty forced us to develop a methodical debugging and validation approach.

4.2 Value delivered to the client

4.2.1 Exceeding quantitative targets

Our solution far exceeded the initial objective:

- Client target: surpass the 86% accuracy of the MATLAB solution
- Result obtained: **93.97% $\pm$ 2.54%** with our DANN architecture
- Improvement: **+7.97 percentage points**

4.2.2 Technical and methodological contributions

Beyond quantitative performance, our work provides several contributions to Dr Zeng:

Optimised DANN architecture:

- Integration of Squeeze-and-Excitation attention mechanisms for 1-D signals
- ResBlocks for hierarchical feature extraction
- Modular, extensible PyTorch framework

Rigorous experimental protocol:

- Cross-validation methodology with five independent runs
- Structured transfer-learning protocol (simulated $\rightarrow$ real domain)
- Comprehensive training-pipeline documentation

4.2.3 Practical research value

Our implementation provides Dr Zeng with a solid foundation for future fault-diagnosis research. The modular code enables extension to other industrial equipment and integration of new domain-adaptation techniques.

4.3 Future work and ideas

4.3.1 Short-term technical improvements

Training optimisation:

- Automated early stopping based on validation metrics
- Exploration of more sophisticated data-augmentation techniques
- Testing alternative feature extractors (Transformer, multi-scale attention)

Robustness and generalisation:

- Validation on other types of industrial robots
- Robustness tests against varying noise levels and variability
- Study of class-imbalance impact on performance

Optimal data-augmentation parameters:

- Generate augmented datasets with different parameters for each class

4.3.2 Methodological extensions

Explainability and interpretability:

- Integrate XAI techniques to understand model decisions
- Visualise features learned by the extractor
- Analyse the importance of different time-series signals in classification

Multi-domain learning:

- Extend learning across multiple robot types simultaneously
- Explore federated learning to pool data
- Develop continual-learning approaches

4.3.3 Industrial applications

Practical deployment:

- Optimise for deployment on embedded systems (edge computing)
- Develop user interfaces for maintenance technicians
- Integrate into CMMS (Computerised Maintenance Management Systems)

Extension to other equipment:

- Adaptation to machine tools, conveyors, handling systems
- Generalisation to rotating equipment (pumps, compressors, fans)
- Application to defect diagnosis in transport systems

4.4 Long-term research outlook

4.4.1 Integration with digital twins

The developed approach could integrate into a broader digital-twin architecture combining:

- Physical modelling of robotic systems
- Prediction of fault evolution over time
- Optimisation of predictive maintenance
- Simulation of fault scenarios for training

4.4.2 Towards autonomous industrial AI

This research fits within a broader vision of smart factories where:

- Diagnostic systems communicate with each other
- Learning occurs continuously from operational data
- Maintenance decisions are automated and optimised
- Adaptation to new operating conditions is autonomous

We would like to express our gratitude to Dr Zhiguo Zeng for his expert supervision and pedagogical approach, which enabled us to develop genuine autonomy in conducting this project.

This experience within the Artificial Intelligence Division perfectly illustrates the value of project-based learning and has given us confidence in our ability to undertake ambitious technical projects. Beyond the technical results obtained, it is above all the formative dimension of this experience that will leave a lasting mark on our engineering careers.

5 Appendices

5.1 Detailed task allocation

Tasks	Victor G.	Victor R.	Anas A.	Mathis J.	Loïc P.
Data analysis	0%	0%	60%	20%	20%
Data augmentation	0%	0%	45%	10%	45%
LSTM development	10%	10%	80%	0%	0%
DANN development	20%	80%	0%	0%	0%
GTN development	0%	0%	0%	20%	80%
HPC management	100%	0%	0%	0%	0%
Experimental validation	20%	20%	20%	20%	20%
Model comparison	20%	20%	20%	20%	20%
Technical documentation	0%	15%	5%	60%	20%
Report writing	20%	20%	20%	20%	20%
Project management	30%	40%	10%	10%	10%

Table 2: Task distribution among team members

6 References

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